Test Cases and Testing Results

Executed evidence for the reconciled local prototype. 26 October 2026.

88 documented cases passed. The wider backend suite passed 151 unittest methods. These are different counting units: one method can contain many parameter cases. The HDB model suite passed 108 tests and Flutter passed 44 tests. Android arm64 debug packaging succeeded. A physical-device run is still pending.

1. Requirement and scope

Lab 4 requirement	Evidence
3.2.1 - one important control class; equivalence classes and boundary values	RecommendationController.recommend: 16 cases. Validated TripRequest: 18 boundary cases. Supporting scoring: 37 cases.
3.2.2 - basis path testing of two complex methods	pricing.window_at and pricing._tier_price. Each graph has V(G)=8 and eight independent feasible paths; one extra full-day condition case is retained.
3.2.3 - carefully minimize cases	One fault or boundary is varied at a time. Shared valid inputs avoid unnecessary combinations. Additional full-day case is justified by an actual mutation.
3.2.4 - execute and record expected versus actual	Machine-readable rows and source hashes: lab4/evidence/test-results.json. Tables below are regenerated from those rows.

Tests use controlled public-code fixtures. No API/database call is needed to reproduce the offline study. Passing fixtures do not establish prediction accuracy, universal tariff coverage or crash freedom on every Android device.

2. Black-box partitions and boundaries

The selected control class coordinates the core UC-01 recommendation. It receives routing and prediction dependencies and returns candidate evidence, exclusions and rankings. The existing recommend(...) function remains a compatibility entry point. Price calculation, collection and scoring stay in focused helpers.

Input/condition	Partitions or boundaries
Forecast outcome	Valid forecast; invalid identity/history; low probability; unsupported horizon; audit failure.
Required evidence	Supported/unsupported price; successful/failed driving or walking route; empty shortlist.
Fallback evidence	Fresh source reading; stale reading; retrieval proxy. A known low forecast cannot enter fallback.
Probability threshold	0.899999 / 0.900000 / 0.900001; tests eligibility, not probability calibration.
Duration, candidates, radius	For each finite integer range, min-1 / min / min+1 / max-1 / max / max+1.
Scoring support	Valid lots/probability modes; negative, non-finite, wrong-type or conflicting availability inputs; invalid weights.

Nominal scoring input is 400 cents, 60 minutes, 500 metres, 900 seconds and one available lot. All penalties equal 0.5, producing score 50. Reference scales and mode weights are policy, not learned driver preferences.

Minimization: 16 controller cases distinguish the application outcomes; 18 request points cover the two bounds of each constrained integer input; 20 equivalence cases and 17 scoring boundary cases provide hand-checkable oracles. The case sets are deliberately not a Cartesian product.

3. Recommendation control cases

ID	Input / change	Expected	Actual / result
CTL-01	valid forecast fixture	('ok', 3, 0)	('ok', 3, 0) | PASS
CTL-02	{'pending': True}	('ok', 3, 0)	('ok', 3, 0) | PASS
CTL-03	{'empty': True}	('no_eligible_candidates', 0, 0)	('no_eligible_candidates', 0, 0) | PASS
CTL-04	{'route_error': 'drive'}	('no_eligible_candidates', 0, 0)	('no_eligible_candidates', 0, 0) | PASS
CTL-05	{'route_error': 'walk'}	('no_eligible_candidates', 0, 0)	('no_eligible_candidates', 0, 0) | PASS
CTL-06	{'price': 'tariff_unverified'}	('no_eligible_candidates', 0, 0)	('no_eligible_candidates', 0, 0) | PASS
CTL-07	{'history_age': 16}	('no_eligible_candidates', 0, 0)	('no_eligible_candidates', 0, 0) | PASS
CTL-08	{'probability': 0.899999}	('no_eligible_candidates', 0, 0)	('no_eligible_candidates', 0, 0) | PASS
CTL-09	{'probability': 0.9}	('ok', 3, 0)	('ok', 3, 0) | PASS
CTL-10	{'probability': 0.900001}	('ok', 3, 0)	('ok', 3, 0) | PASS
CTL-11	{'audit_error': True}	('no_eligible_candidates', 0, 0)	('no_eligible_candidates', 0, 0) | PASS
CTL-12	{'prediction_status': ' unsupported_horizon_15_to_90_minutes'}	('current_reading_fallback', 0, 3)	('current_reading_fallback', 0, 3) | PASS
CTL-13	{'prediction_status': ' unsupported_horizon_15_to_90_minutes', 'current_status': ' stale'}	('no_eligible_candidates', 0, 0)	('no_eligible_candidates', 0, 0) | PASS
CTL-14	{'current_count': 0}	('ok', 3, 0)	('ok', 3, 0) | PASS
CTL-15	{'wrong_identity': True}	('no_eligible_candidates', 0, 0)	('no_eligible_candidates', 0, 0) | PASS
CTL-16	{'prediction_status': ' unknown_carpark', ' timestamp_kind': 'retrieval'}	('no_eligible_candidates', 0, 0)	('no_eligible_candidates', 0, 0) | PASS

4. Request boundary values

ID	Input / change	Expected	Actual / result
REQ-01	duration_minutes=0; other fields nominal	ValueError	ValueError | PASS
REQ-02	duration_minutes=1; other fields nominal	accepted	accepted | PASS
REQ-03	duration_minutes=2; other fields nominal	accepted	accepted | PASS
REQ-04	duration_minutes=1439; other fields nominal	accepted	accepted | PASS
REQ-05	duration_minutes=1440; other fields nominal	accepted	accepted | PASS
REQ-06	duration_minutes=1441; other fields nominal	ValueError	ValueError | PASS
REQ-07	candidate_limit=0; other fields nominal	ValueError	ValueError | PASS
REQ-08	candidate_limit=1; other fields nominal	accepted	accepted | PASS
REQ-09	candidate_limit=2; other fields nominal	accepted	accepted | PASS
REQ-10	candidate_limit=4; other fields nominal	accepted	accepted | PASS
REQ-11	candidate_limit=5; other fields nominal	accepted	accepted | PASS
REQ-12	candidate_limit=6; other fields nominal	ValueError	ValueError | PASS
REQ-13	radius_meters=99; other fields nominal	ValueError	ValueError | PASS
REQ-14	radius_meters=100; other fields nominal	accepted	accepted | PASS
REQ-15	radius_meters=101; other fields nominal	accepted	accepted | PASS
REQ-16	radius_meters=2999; other fields nominal	accepted	accepted | PASS
REQ-17	radius_meters=3000; other fields nominal	accepted	accepted | PASS
REQ-18	radius_meters=3001; other fields nominal	ValueError	ValueError | PASS

5. Supporting scoring equivalence classes

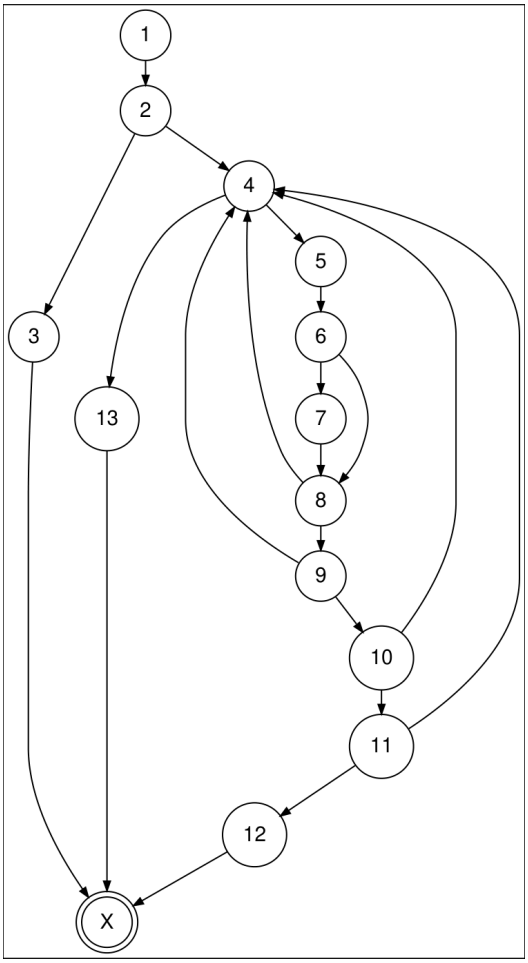
ID	Input / change	Expected	Actual / result
EC-01	nominal	50.0 (best_overall_v1_provisional)	50.0 (best_overall_v1_provisional) | PASS
EC-02	available_lots=None, p_free=0.5, weights={'cost': 0.25, 'walking': 0.25, 'availability': 0.25, 'driving': 0.25}	50.0 (best_overall_probability_v2_provisional)	50.0 (best_overall_probability_v2_provisional) | PASS
EC-03	cost_cents=-400	ValueError	ValueError | PASS
EC-04	cost_cents=400.0	ValueError	ValueError | PASS
EC-05	duration_minutes=-60	ValueError	ValueError | PASS
EC-06	duration_minutes=60.5	ValueError	ValueError | PASS
EC-07	walking_meters=-500	ValueError	ValueError | PASS
EC-08	walking_meters=inf	ValueError	ValueError | PASS
EC-09	walking_meters='500'	ValueError	ValueError | PASS
EC-10	driving_seconds=-900	ValueError	ValueError | PASS
EC-11	driving_seconds=nan	ValueError	ValueError | PASS
EC-12	available_lots=-5	ValueError	ValueError | PASS
EC-13	available_lots=None	ValueError	ValueError | PASS
EC-14	p_free=0.5	ValueError	ValueError | PASS
EC-15	available_lots=None, p_free=-0.5	ValueError	ValueError | PASS
EC-16	available_lots=None, p_free=1.5	ValueError	ValueError | PASS
EC-17	available_lots=None, p_free=nan	ValueError	ValueError | PASS
EC-18	weights={'cost': 0.5, 'walking': 0.3, 'availability': 0.2}	ValueError	ValueError | PASS
EC-19	weights={'cost': 0.5, 'walking': 0.5, 'availability': 0.0, 'driving': 0.0}	ValueError	ValueError | PASS
EC-20	weights={'cost': 0.5, 'walking': 0.5, 'availability': 0.5, 'driving': 0.5}	ValueError	ValueError | PASS

6. Supporting scoring boundary values

ID	Input / change	Expected	Actual / result
BV-01	cost_cents=-1	ValueError	ValueError | PASS
BV-02	cost_cents=0	70.0	70.0 | PASS
BV-03	cost_cents=1	69.9002	69.9002 | PASS
BV-04	duration_minutes=0	ValueError	ValueError | PASS
BV-05	duration_minutes=1	30.6557	30.6557 | PASS
BV-06	walking_meters=-0.001	ValueError	ValueError | PASS
BV-07	walking_meters=0	65.0	65.0 | PASS
BV-08	driving_seconds=-0.001	ValueError	ValueError | PASS
BV-09	driving_seconds=0	55.0	55.0 | PASS
BV-10	available_lots=-0.001	ValueError	ValueError | PASS
BV-11	available_lots=0	40.0	40.0 | PASS
BV-12	available_lots=None, p_free=-0.001	ValueError	ValueError | PASS
BV-13	available_lots=None, p_free=0	40.0	40.0 | PASS
BV-14	available_lots=None, p_free=1	60.0	60.0 | PASS
BV-15	available_lots=None, p_free=1.001	ValueError	ValueError | PASS
BV-16	weights={'cost': 0.1, 'walking': 0.2, 'availability': 0.3, 'driving': 0.4}	50.0	50.0 | PASS
BV-17	weights={'cost': 0.1, 'walking': 0.2, 'availability': 0.3, 'driving': 0.400001}	ValueError	ValueError | PASS

7. Basis path study: window_at

N=14, E=20; V(G)=E-N+2=8. Seven statement-level decisions plus one also gives eight. Boolean operands inside a compound condition are not separate nodes in this graph; this is not an MC/DC claim.



1	Read rule configuration and clocks
2	Equal clocks without full_day?
3	Reject empty window
4	Next anchor date / exhausted
5	Construct dated start and end
6	Closing <= opening?
7	Move end into next day
8	Minute outside interval?
9	Day category incompatible?
10	Weekday filter excludes date?
11	Holiday-eve filter mismatch?
12	Return interval
13	Return no match
X	Shared exit

X is the shared virtual exit. Edge direction follows the source; true/false outcomes are specified in the path sequences below. Each declared path was checked against an actual line trace.

The eight basis trace vectors have rank eight over GF(2), using edge traversal parity. That verifies independence for this selected graph; it does not mean all possible paths were executed.

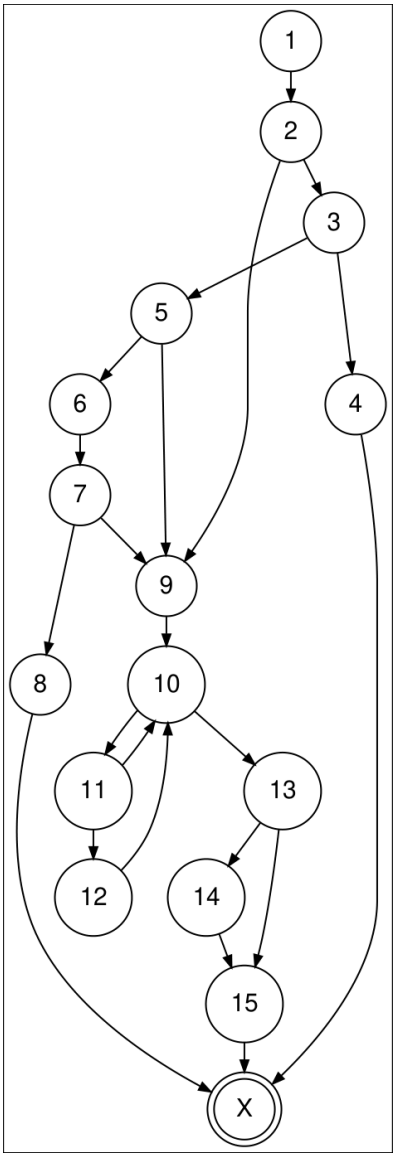
window_at: executed paths and outputs

ID	Path	Input	Expected / actual
WA-P1	1-2F-4-5-6F-8F-9F-10F-11F-12	weekday 07:00-17:00; at Mon 10:00	Mon 07:00 to Mon 17:00 / Mon 07:00 to Mon 17:00
WA-P2	1-2T-3	all 07:00-07:00; at Mon 10:00	None / None
WA-P3	1-2F-4-5-6T-7-8F-9F-10F-11F-12	all 19:00-07:00; at Mon 20:00	Mon 19:00 to Tue 07:00 / Mon 19:00 to Tue 07:00
WA-P4	1-2F-4-5-6F-8T-4-5-6F-8T-4-13	weekday 07:00-17:00; at Mon 18:00	None / None
WA-P5	1-2F-4-5-6F-8F-9T-4-5-6F-8T-4-13	weekday 07:00-17:00; at Sat 10:00	None / None
WA-P6	1-2F-4-5-6F-8F-9F-10T-4-5-6F-8T-4-13	all 07:00-17:00, {'weekdays': [0, 1, 2, 3, 4]}; at Sat 10:00	None / None
WA-P7	1-2F-4-5-6F-8F-9F-10F-11T-4-5-6F-8T-4-13	all 07:00-17:00, {'ph_eve': True}; at Mon 10:00	None / None
WA-P8	1-2F-4-5-6T-7-8T-4-5-6T-7-8F-9F-10F-11F-12	all 19:00-07:00; at Tue 02:00	Mon 19:00 to Tue 07:00 / Mon 19:00 to Tue 07:00
WA-C1	1-2F-4-5-6T-7-8F-9F-10F-11F-12	all 07:00-07:00, {'full_day': True}; at Mon 10:00	Mon 07:00 to Tue 07:00 / Mon 07:00 to Tue 07:00

WA-C1 is additional condition coverage, not a ninth basis path. Changing closing <= opening to closing < opening leaves the eight basis cases passing but fails WA-C1. The case checks that equal opening/closing times mean 24 hours only when full_day is explicit.

8. Basis path study: _tier_price

N=16, E=22; V(G)=E-N+2=8. Seven statement-level decisions plus one also gives eight. Boolean operands inside a compound condition are not separate nodes in this graph; this is not an MC/DC claim.



1	Read rule configuration and tiers
2	First-entry status required?
3	Status unknown?
4	Return unknown
5	Repeat entry?
6	Read repeat-entry tiers
7	Repeat tiers missing?
8	Return unknown
9	Initialize amount and remaining
10	Next tier / exhausted
11	Used minutes > 0?
12	Charge and consume tier minutes
13	Minutes remain?
14	Round up remainder charge
15	Return exact amount
X	Shared exit

X is the shared virtual exit. Edge direction follows the source; true/false outcomes are specified in the path sequences below. Each declared path was checked against an actual line trace.

The eight basis trace vectors have rank eight over GF(2), using edge traversal parity. That verifies independence for this selected graph; it does not mean all possible paths were executed.

_tier_price: executed paths and outputs

ID	Path	Input	Expected / actual
TP-P1	1-2F-9-10-11T-12-10-13T-14-15	tiers [60m@120c/60m]; 90 min; first_entry=None	170 / 170
TP-P2	1-2T-3T-4	tiers [60m@120c/60m]; first entry required; 90 min; first_entry=None	None / None
TP-P3	1-2T-3F-5F-9-10-11T-12-10-13T-14-15	tiers [60m@120c/60m]; first entry required; 90 min; first_entry=True	170 / 170
TP-P4	1-2T-3F-5T-6-7T-8	tiers [60m@120c/60m]; first entry required; 90 min; first_entry=False	None / None
TP-P5	1-2T-3F-5T-6-7F-9-10-11T-12-10-13T-14-15	tiers [60m@120c/60m]; first entry required; repeat tier 60m@200c/60m; 90 min; first_entry=False	250 / 250
TP-P6	1-2F-9-10-13T-14-15	tiers []; 45 min; first_entry=None	100 / 100
TP-P7	1-2F-9-10-11T-12-10-11F-10-13F-15	tiers [60m@120c/60m, 60m@80c/30m]; 60 min; first_entry=None	120 / 120
TP-P8	1-2F-9-10-11T-12-10-13F-15	tiers [60m@120c/60m]; 60 min; first_entry=None	120 / 120

9. Regression, reproducibility and limits

Check	Observed result / scope
Backend regression	151 unittest methods, zero failures/errors/skips. Includes the merged operator tests and ranking modes.
HDB model tests	108 tests passed; selected unit/service suites. Ignored raw-data parity archives were not supplied.
Flutter	44 tests passed; analyzer clean. Includes operator entry, response validation, lifecycle/stress, map and navigation fixtures.
Android build	arm64 debug APK built with API URL http://127.0.0.1:8000. USB reverse is required for this device URL.
Live integration	See evidence/live-check.json. This check observes current providers; it is not a later-outcome accuracy evaluation.
Physical phone	No Android device was connected during this preparation. Do not mark the checklist below passed without execution.

Reproduce the documented study: python lab4/collect_evidence.py using the backend Python environment. It evaluates each table row, traces both methods, checks graph complexity/independence and runs backend regression. Source SHA-256 hashes and timestamps are stored with the results.

Known scope: HDB short-mode forecasts only, 15-90-minute integration horizon, 0.90 ranking threshold; selected model evaluation horizons were 15/30/45/60/90. Current fallback is separate. Pricing is conditional on fresh supported rules, and shortlisted carparks are not a global optimum. Operator console actions, durable sessions and several local-data features remain prototypes.

10. Android acceptance record - pending

ID	Action	Expected	Actual status
AND-01	Install the supplied arm64 debug APK on the actual demo phone.	Launches anonymous Search.	Pending on device
AND-02	Use USB reverse port 8000; issue a real search.	Phone reaches the laptop backend.	Pending on device
AND-03	Grant foreground location; then test denial/manual origin.	Fresh fix or recoverable manual-input path.	Pending on device
AND-04	Search a rehearsed 15-90-minute arrival scenario.	Real routed prices and eligible arrival probabilities, or an honest unavailable state.	Pending on device
AND-05	Re-search with each ranking mode.	Mode sent; old results cleared; policy shown consistently.	Pending on device
AND-06	Open map, launch navigation and return to the app.	Attributed tiles/markers and correct carpark handoff; no crash.	Pending on device
AND-07	Run a short trip and inspect current-only fallback.	Fallback is separately labelled; no fabricated forecast.	Pending on device
AND-08	Rotate, open keyboard, background/foreground, switch tabs repeatedly.	No crash/overflow; transient resets understood.	Pending on device
AND-09	Disconnect/reconnect USB and rerun reverse; retry a request.	Recoverable service error and successful retry.	Pending on device
AND-10	If showing Admin, test wrong/right configured operator credentials.	Credential check works; console data/actions identified as prototypes.	Pending on device

For each completed phone check, record device model, Android version, APK hash, date, observed result and tester identity in the team copy. The build and widget tests do not substitute for this record.